CWV-Net: A Deep Neural Network for Atmospheric Column Water Vapor Retrieval From Hyperspectral VNIR Data

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***Abstract*— Estimation of the total column water vapor (CWV) content of the atmosphere plays an important role in the atmospheric compensation (AC) of remotely sensed hyperspectral images collected in the visible and near infrared (VNIR) spectral range. Most of the proposed CWV retrieval methods provide accurate estimates as long as other significant atmospheric parameters are known. Those parameters are not generally available and must in turn be estimated. In this article, a new approach based on deep learning is proposed that allows the estimation of CWV without the knowledge of the atmospheric visibility, the solar zenith angle, and the atmospheric point spread function (PSF). The proposed approach includes a training strategy based on synthetic data that are generated according to an accurate radiative-transfer model, and by exploiting reflectance spectral libraries and the MODTRAN radiative-transfer code. Experiments on simulated data are carried out to analyze the performance of the proposed deep neural network with reference to both aerial and satellite applications. Furthermore, an example of the results provided by the network in a real application is shown. For this purpose, the network is applied to data acquired by an airborne hyperspectral sensor operating in the VNIR spectral range.**

***Index Terms*— Atmospheric compensation (AC), hyperspectral imagery, water vapor retrieval.**

I. Introduction

HYPERSPECTRAL sensors (HSs) collecting data in the visible and near infrared (VNIR) spectral range have gained the interest of the scientific community in the last years, thanks to the recent development of new generation sensors operating on board of unmanned aerial vehicle (UAV) and drones (see [1]–[3]). The major advantage offered by hyperspectral technology with respect to other remote-sensing systems is the possibility of exploiting the physical and chemical compositions of the observed surface through its spectral reflectance. Unfortunately, due to the distortion introduced by the atmosphere in the radiative transfer path, HSs do not measure directly the spectral reflectance of the monitored surface but rather the at-sensor radiance. Thus, the acquired data must be processed to remove from the measured at-sensor radiance the contributions due to the solar illumination and the atmospheric effects. Such a process is generally referred to as an atmospheric compensation (AC, see [4], [5]) and plays a crucial role in the exploitation of hyperspectral data. AC is a very challenging task as the effect of solar illumination and the distortion induced by the atmosphere depends on several factors, mainly aerosol scattering and water vapor absorption, many of which are not measurable at the acquisition time in most of the practical applications.

Among the numerous AC techniques proposed in the past, the most accurate and reliable are the physics-based ones that exploit analytical models of the radiative transfer (radiative transfer models(RTMs), see [6]–[9]) to account for the absorption and scattering from the atmospheric components and the acquisition geometry. Physics-based (or RTM-based) AC methods generally deal with two distinct issues (see [10]): 1) estimation of the atmospheric parameters of the RTM and 2) inversion of the model for spectral reflectance retrieval.

Usually, the parameters of the RTM related to the acquisition geometry are obtained from the mission planning or can be derived from the on-board instrumentation. The remaining parameters can be only estimated by the observed data. For the VNIR spectral range, the relevant RTM parameters are the meteorological range or atmospheric visibility (V) and the total column water vapor (CWV) concentration of the atmosphere. V can be reasonably considered constant all over the scene and is generally estimated by analyzing specific regions of the acquired image using methods such as the dense dark vegetation (DDV, see [11]), the boundary shaded pixels algorithm (see [9]), or by exploiting a reference (Lambertian) target deployed in the monitored scene with known spectral reflectance.

The CWV concentration is generally spatially variant and its estimation needs a pixel-level approach.

Several CWV retrieval algorithms resort to the differential absorption approach. For instance, such an approach is used by the AC commercial software package named fast line-of-sight atmospheric analysis of spectral hypercubes (FLAASH, see [12]). One of the most popular CWV retrieval algorithm is the atmospheric precorrected differential absorption technique (APDA) proposed in [13], and included in the airborne

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atmospheric and topographic correction (ATCOR, see [8]) software package.

APDA makes use of the at-sensor radiance values in three groups of channels: the measurement channels with spectral central wavelength falling within a given atmospheric water vapor absorption region (typically around 820 or 940 nm) and two "reference" channels corresponding to the wavelengths located at each side of the water absorption band where absorption due to CWV is actually negligible. Those at-sensor radiance data are combined to compute the "depth" of the absorption feature that is related to CWV. The computed "depth" is compared with a precalculated lookup table (LUT) to obtain the estimate of the CWV parameter.

APDA makes use of two simplifying assumptions: 1) the effects of the radiation of the surfaces surrounding the considered pixel (adjacency effect) are neglected and 2) the value of the spectral reflectance in the measurement channel is set to a fixed value (0.4) regardless of the observed surface. Both the hypotheses are not realistic. It is well known that adjacency effects are not negligible in real hyperspectral data (see [8], [14]). Furthermore, the spectral reflectance in the measurement channel strongly depends on the observed surface. Recently (see [16]), by analyzing the performance of APDA using a more realistic and well-established RTM (see [10]), it has been shown that the impact of such hypotheses is not negligible. To overcome the drawbacks of the APDA, in [16], the low-rank subspace projection-based water estimator (LRSWE) was proposed. LRSWE is a CWV retrieval algorithm that exploits a more realistic RTM accounting for the adjacency effects and addresses the variability of the spectral reflectance of the observed surface in the water absorption bands by resorting to a low-rank subspace representation. In LRSWE, CWV estimates are obtained by solving a nonlinear least square optimization problem making use of specific LUT derived by the MODTRAN (see [17]) radiative-transfer code.

LRSWE outperforms APDA in CWV retrieval even in the presence of noise. However, the practical application of the algorithm has some critical aspects. First, the LUT adopted by the algorithm (as well as that used by APDA) are derived assuming known some RTM parameters such as the flight height h, the solar zenith angle ϑ, and the atmospheric visibility V. In general, h is obtained with high accuracy from the flight operation planning; ϑ can be derived from the acquisition date and time, while V is estimated by image-based algorithms (such as those cited above) and represents the main source of uncertainty in AC. Of course, errors in the values assigned to the three parameters negatively affect CWV estimation performance. Second, to account for the adjacency effects, LRSWE makes use of the adjacent at-sensor radiance that is obtained as a linear combination of the radiances of a neighborhood of the pixel under consideration. The weights adopted in the linear combination should be set according to the atmospheric point spread function (PSF) (see [18]) that is not generally known. Since PSF estimation is a very challenging task, it is a common practice in hyperspectral AC (see [8], [19]) to refer to standard PSF models by introducing an uncertainty in the derivation of the adjacent at-sensor radiance.

This has motivated us to develop a new strategy to provide accurate estimates for CWV regardless of the perfect knowledge of all the parameters of the RTM (specifically V and ϑ) and the atmospheric PSF. To obtain such a general algorithm, we exploit the potential of the deep learning (DL).

DL allows models with increasingly higher semantic layers and complexity to be derived from the data themselves through a learning process. This makes DL a promising approach to achieve high precision and flexibility in solving very complex problems. DL has shown to yield remarkable results in a large number of applications concerning image analysis and speech processing. In the last few years, DL has been successfully applied to hyperspectral images to perform various tasks (see [20], [21]) mainly concerning classification, feature extraction, and noise mitigation. DL classification algorithms exploiting the spectral features of hyperspectral pixels were proposed in [22] and [23], while DL approaches based on the exploitation of the spatial features can be found in [24] and [25]. Recently, DL algorithms able to analyze both spectral and spatial features have been proven to enhance the classification accuracy (see [26]). DL techniques, such as stacked autoencoders and convolutional neural networks (CNNs), are also used for spectral-spatial feature extraction (see [27], [28]) and image denoising (see [29], [30]). To the best of our knowledge, no work has been yet published concerning the application of DL techniques to the estimation of RTM parameters.

In this article, we investigate the application of deep neural networks (DNNs) to CWV retrieval. Specifically, we analyze one-dimensional-CNN (1D-CNN)-based architecture named CWV-Net for estimating the total CVW concentration of the atmosphere in the VNIR spectral range from each pixel of a given hyperspectral image. CWV-Net is designed and trained to provide CWV estimates, without specific knowledge about the solar zenith angle and the atmospheric visibility. The only parameter to be set is the flight height h. Training is performed over hyperspectral data synthetized by means of a simulation strategy powered by the MODTRAN radiative-transfer code and exploiting the availability of spectral reflectance libraries.

This article is organized as follows. In Section II, we briefly summarize the RTM, while in Section III, we introduce the network architecture. In Section IV, we describe the strategy adopted to train the network by detailing the methodology for simulating the training data. Section V is dedicated to an extensive analysis of the performance provided by the proposed DL-based algorithm on simulated data together with the comparison between CWV-Net and the LRSWE algorithm. We conclude this article with a proof of concept concerning the application of CWV-Net to a real hyperspectral image.

II. RTM for VNIR Hyperspectral Images

In this section, we introduce the RTM that is generally adopted (see [8], [9]) to characterize the radiative transfer in the atmosphere in the VNIR spectral range. The RTM is a fundamental part of the proposed DL-based estimation strategy as it is used in the construction of the training set adopted in the network learning.

Let us start by denoting as $L(\lambda)$ the at-sensor spectral radiance from the observed ground resolution cell in the sensor

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channel with central wavelength λ. It can be expressed as

$$L(\lambda) = L_{\mathrm{GND}}(\lambda, \mathbf{\Gamma}, \mathrm{CWV}) + L_p(\lambda, \mathbf{\Gamma}, \mathrm{CWV}) \quad (1)$$

where $L_p(\lambda, \mathbf{\Gamma}, \mathrm{CWV})$ is the intrinsic upwelling atmospheric spectral radiance and $L_{\mathrm{GND}}(\lambda, \mathbf{\Gamma}, \mathrm{CWV})$ is the spectral radiance reaching the sensor after the interaction with the ground surface (ground reflected radiance). Both depend on the parameters specifying the acquisition geometry, the effects of absorption by gases in the atmosphere, and scattering by molecules and aerosol. In general, the acquisition geometry parameters include the sun zenith (ϑ) and azimuth angles, the sensor zenith, and azimuth angles, the sensor altitude (h), and the terrain elevation. In this article we assume that data are acquired from nadir view and that the monitored scene is flat. Thus, the relevant parameters for acquisition geometry characterization are ϑ and h.

The scattering effects are defined by the aerosol type and the atmospheric visibility (V). The aerosol-type depends on the geographical position of the monitored scene, on the seasonal characteristics at the acquisition time, and, in general, on the concentration of the various contents of the atmosphere. The absorption effects depend on the concentration of the gases in the atmosphere. In the VNIR, spectral range absorption is determined by the oxygen, which affects very narrow portions of the electromagnetic spectrum, by the ozone, whose effects are well modeled by standard concentration values, and mainly by the water vapor which is characterized by the total CWV concentration. In (1), we decoupled the parameter CWV by the rest of the atmospheric model parameters that are grouped in the vector $\mathbf{\Gamma}$.

The ground reflected radiance $L_{\mathrm{GND}}(\lambda, \mathbf{\Gamma}, \mathrm{CWV})$ is generally modeled as (see [8], [9], [8], [14])

$$\begin{aligned} &L_{\mathrm{GND}}(\lambda, \mathbf{\Gamma}, \mathrm{CWV}) \\ &\quad = \frac{E_g(\lambda)}{\pi}\left[\tau_1(\lambda, \mathbf{\Gamma}, \mathrm{CWV})\frac{\rho(\lambda)}{1 - S(\lambda, \mathbf{\Gamma}, \mathrm{CWV})\rho_a(\lambda)}\right. \\ &\qquad + \tau_2(\lambda, \mathbf{\Gamma}, \mathrm{CWV}) \\ &\qquad \left.\frac{\rho_a(\lambda)}{1 - S(\lambda, \mathbf{\Gamma}, \mathrm{CWV})\rho_a(\lambda)}\right]. \end{aligned} \quad (2)$$

In (2), $E_g(\lambda)$ is the solar spectral irradiance incident on the ground, $\rho(\lambda)$ and $\rho_a(\lambda)$ are the spectral reflectances of the observed surface and of the adjacent surfaces, respectively. The term $\tau_1(\lambda, \mathbf{\Gamma}, \mathrm{CWV})$ accounts for the interaction of the surface material with the direct and diffuse solar radiation, while $\tau_2(\lambda, \mathbf{\Gamma}, \mathrm{CWV})$ accounts for the part of the solar radiation that, after the interaction with the materials in a neighborhood of the surface of interest, is scattered by the atmosphere in the sensor instantaneous field of view (IFOV). Finally, the contributions due to multiple scattering are addressed by the term $1 - S(\lambda, \mathbf{\Gamma}, \mathrm{CWV})\rho_a(\lambda)$ with $S(\lambda, \mathbf{\Gamma}, \mathrm{CWV})$ denoting the spectral spherical albedo of the atmosphere.

The RTM in (1) and (2) refer to a Lambertian observed surface. Thus, $\rho(\lambda)$ is the isotropic spectral reflectance usually named hemispherical-directional reflectance factor (HDRF, see [31]). In general, $\rho(\lambda)$ can be viewed as a scaled version of the "absolute" surface reflectance (see [4]) in that it does not address the anisotropic reflective behavior of the surface.

TABLE I

PARAMETERS OF THE CONVOLUTIONAL LAYERS: NUMBER OF KERNELS K_l, SIZE OF EACH KERNEL T_l, MAX-POOLING DIMENSION P_l, AND STRIDE S_l

1D-CNN layer	K_l	T_l	P_l	S_l
1	4	4	4	4
2	16	4	4	4
3	32	4	4	4
4	64	1	3	3

Furthermore, it is worth noting that the presented model refers to clear sky condition that is the desirable condition for hyperspectral data collection.

III. CWV-NET ARCHITECTURE

The goal of CWV-Net is the estimation of the atmospheric parameter CWV in each pixel of the hyperspectral image when one has no knowledge of the atmospheric visibility (V), of the solar zenith angle (ϑ) and of the atmospheric PSF. Only the sensor altitude h and the generic knowledge of the parameters related to aerosol scattering (that will be detailed in the next section) are required. To accomplish the above-cited estimation task, we resort to the potential of DNNs.

DNNs have several hidden layers and have been shown to be able to represent complicated data structures. A DNN takes advantages from its multilevels architecture to obtain hierarchical feature representation of the data of interest. The parameters of each layer of a DNN are generally initialized randomly and finely tuned in a supervised manner by using labeled data and an error back-propagation algorithm. The number of the parameters to be learned in a DNN is very high and grows with the number of hidden layers. This is one of the factors that has limited the use of DNNs in the past. The huge number of DNN's parameters along with the limited number of available training data cause overfitting and a low capability of generalization. These issues are partially solved by the introduction of the CNN architecture (see [32]) that exploits the ideas of local connections and shared weights within the hidden layers to obtain deep networks with a number of parameters smaller than those characterizing the DNNs based on dense layers.

Since we are interested into estimating CWV from the at-sensor spectral radiance measured by a HS, in this article we focus on the class of 1D-CNN-based DNN for regression. Such class of DNN have a general common architecture comprising.

1) One input layer.
2) One or more 1D-CNN hidden layers that are in charge of extracting hierarchically the significant features from the data.
3) One or more fully connected layers [multilayer perceptrons (MLP)] that, by exploiting the extracted features, estimate the parameter of interest.

The generic lth 1D-CNN layer works on the features extracted by the previous layer and provides a new set of features. It has a fixed number K_l of kernels (or filters) having size $T_l \times 1$, that slide (convolve) along the layer input data dimension. The

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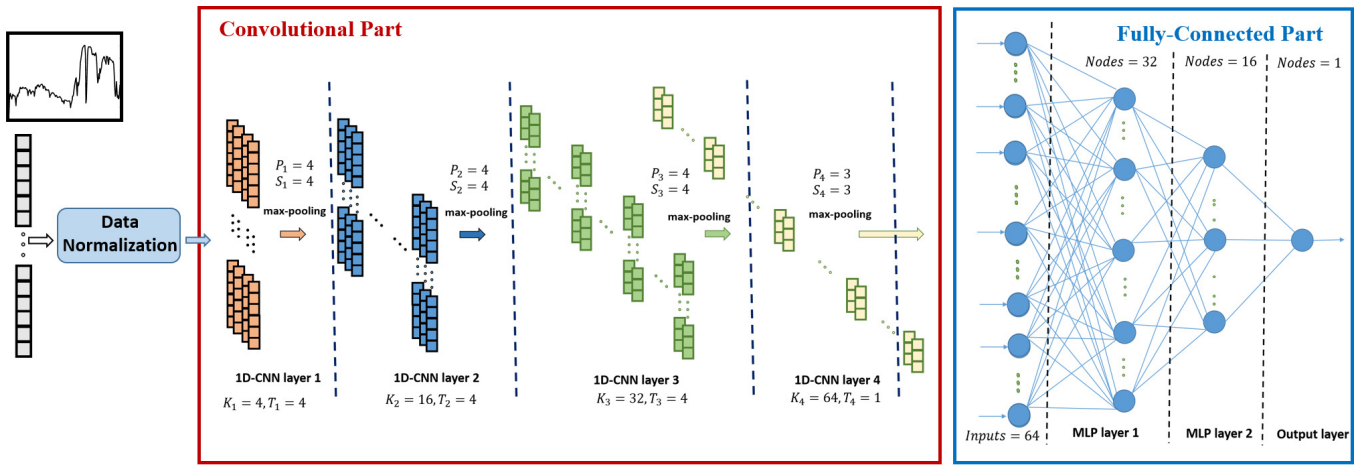


Fig. 1. CWV-Net architecture.

outputs of each kernel are modified by the activation function (which is generally nonlinear). Finally, down sampling is performed on the output data by the pooling layer. It combines $P_l \times 1$ patches of its input data vector providing a single feature for each patch. In general, such a feature is obtained by taking the maximum (max-pooling) or averaging (average-pooling) over the patch values. The patches are selected sequentially over the data vector with a step of S_l (stride). It is noted that pooling can offer robustness to local features shift.

The proposed CWV-Net complies with the general architecture summarized above. The input vector for CWV-Net is the spectral radiance measured by the adopted HS. Thus, the number of sensor channels defines the nodes of the input layer.

In this article, we refer to the "Sistema Iperspettrale Modulare. Galileo Avionica" (SIM.GA) sensor, an avionic hyperspectral instrument manufactured by Leonardo–Finmeccanica covering the spectral range from 400 to 1000 nm with a spectral sampling of 1.2 nm. In general, data acquired by the sensor are preprocessed by binning adjacent channels with a factor of two to reduce the effect of random noise. The resulting number of spectral channels is equal to 256. It is worth noting that the CWV estimation over vegetated pixels is generally biased by the leaf water content. It was shown that the water absorption features dominate the spectral reflectance of vegetation in the near-infrared region of the spectrum (see [33]). Specifically, the first significant absorption can be approximately located from 925 to 1000 nm (see [33], [34]). Thus, to avoid leaf water content bias over vegetated areas, we limit the spectral radiance provided by the sensor to 925 nm. The resulting number of spectral bands and, therefore, the number of nodes of the input layer of CWV-Net is 228.

The convolutional part of the proposed DNN has four 1D-CNN layers. In all of them, the activation function is the rectified linear unit (ReLU) and down-sampling is obtained through max-pooling. The number and the size of the kernels along with the max-pooling dimension and stride in each 1D-CNN layer are summarized in Table I. The output of the convolutional part is a 64×1 feature vector.

The rest of CWV-Net comprises two fully connected hidden layers having 32 and 16 nodes, respectively. The output layer has one single node being CWV-Net conceived for the estimation of a scalar parameter. The fully connected layers and the output layer have a ReLU activation function.

The complete architecture of CWV-Net is summarized in Fig. 1. It is worth noting that the input of the DNN is preprocessed to have unit Euclidean norm ("Data Normalization").

IV. CWV-Net Training

The learning phase of CWV-Net is performed by minimizing a given loss function through an error back-propagation algorithm over a training set.

The training set is obtained by simulation exploiting MODTRAN and a spectral reflectance library (**D**) obtained merging three different spectral libraries: ASTER (see [35]), USGS (version released in 2007, see [36]), and ANGERS (see [37]) that contain the measured spectral reflectances of several man-made materials, vegetation, water, soils, and minerals.

In this section, we describe the simulation strategy adopted to construct the training set and we introduce the loss function minimized in the learning stage.

A. Training Set Simulation Strategy

The tth element of the training set is a pair comprising a value CWV_t of the CWV concentration and the corresponding at-sensor spectral radiance vector $\boldsymbol{L}_t(\mathrm{CWV}_t)$. The element of the vectors are the values of the spectral radiance corresponding to the central wavelength of each band of the considered HS.

$\boldsymbol{L}_t(\mathrm{CWV}_t)$ is generated according to the RTM in (1) and (2) by means of the simulation strategy summarized in Fig. 2.

Such a strategy comprises three main blocks. The first is the reflectance generator block that simulates the surface spectral reflectance vector $\boldsymbol{\rho}_t$ and the adjacent spectral reflectance vector $\boldsymbol{\rho}_{a,t}$ from the elements of the library **D**. To increase the number and the variability of the spectral reflectance samples,,

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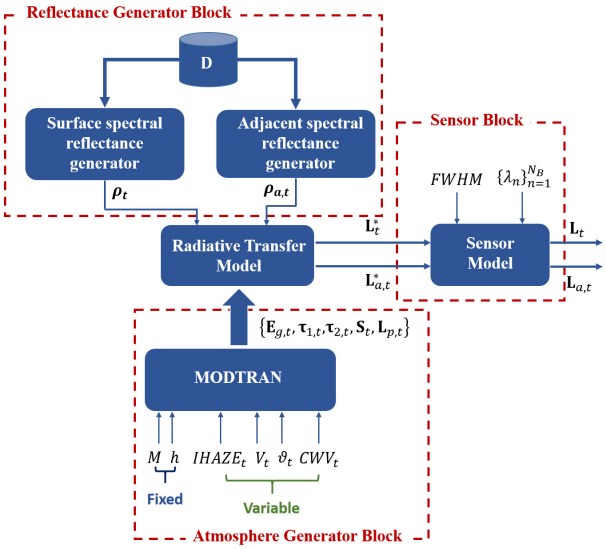


Fig. 2. Block diagram of the simulation strategy.

we use a data augmentation strategy based on the linear mixing model (see [38]). Specifically

$$\boldsymbol{\rho}_t = \sum_{i=1}^{5} a_i \mathbf{d}_i \tag{3}$$

where $\mathbf{d}_i$ with $i = 1, \ldots, 5$ are the elements of $\mathbf{D}$ that acts as endmembers, and a_i with $i = 1, \ldots, 5$ are the abundances. The endmembers are selected randomly from $\mathbf{D}$ and the abundances are drawn from a Dirichlet distribution so as to meet the sum-to-one and the nonnegative constraints. The same model is adopted to simulate $\boldsymbol{\rho}_{a,t}$, using new randomly selected endmembers and new randomly generated abundances. It is worth noting that the use of the linear mixing model is a data augmentation strategy introduced to avoid overfitting over the spectra of the dictionary D, and it is neither a requirement nor a fundamental hypothesis for the correct functioning of the proposed CWV retrieval algorithm.

The atmosphere generator block is based on MODTRAN and generates the spectral vector of the radiometric quantities $\boldsymbol{E}_{g,t}$, $\boldsymbol{\tau}_{1,t}$, $\boldsymbol{\tau}_{2,t}$, $\boldsymbol{L}_{p,t}$, and $\boldsymbol{S}_t$ that appears in the RTM of (1) and (2). They are generated for a given set of the atmospheric model parameters required by MODTRAN: M, IHAZE_t, h, V_t, ϑ_t and, of course, CWV_t. The parameters M (called MODEL in MODTRAN) and IHAZE_t specify the geographical and seasonal model and the aerosol model (see [39]), respectively. The parameters h, V_t, and ϑ_t (already introduced in Sections I and II) represent the sensor height, the atmospheric visibility, and the solar zenith angle.

The radiometric quantities and the reflectances $\boldsymbol{\rho}_t$ and $\boldsymbol{\rho}_{a,t}$, are combined according to the RTM defined by (1) and (2) to obtain the at-sensor spectral radiance vector $\mathbf{L}_t^*$ that is transformed in the sensor domain ($\mathbf{L}_t$) by the Sensor Block. It accounts for the spectral response functions (SRFs) of the N_B sensor channels. Usually, a Gaussian-shaped SRF is considered for each channel, it is characterized by the full width half maximum (FWHM) and the channel central wavelength (λ_n). In the case of the SIM.GA sensor, the FWHM is 2.284 nm for each channel.

TABLE II
VALUES ADOPTED FOR THE VARIABLE ATMOSPHERIC PARAMETERS IN THE CONSTRUCTION OF THE TRAINING SET

Variable Atmospheric Parameters	
$IHAZE_t$	1 (Rural); 3 (Navy Maritime); 4 (Maritime); 5 (Urban); 10 (Desert)
$\vartheta_t(deg)$	0:10:70
$V_t(Km)$	10:10:60, 90, 120
$CWV_t\ (^{g}/_{cm^2})$	0.5:0.1:5

To construct the training set M and h are kept constant while the other parameters are made to vary. The values adopted for the variable parameters are shown in Table II.

For a value $\overline{CWV}$ of the total CWV concentration in Table II, 1000 different at-sensor spectral vectors $\mathbf{L}_t$ are generated according to the proposed simulation strategy and selecting randomly the other variable atmospheric parameters from the values in Table II. Thus, we obtain 1000 pairs $\left\{\mathbf{L}_t\left(\overline{CWV}\right), \text{CWV}_t = \overline{CWV}\right\}$. The procedure is repeated for each of the 46 values of the total CWV concentration in Table II, thus obtaining a total number of training pairs equal to 46000. Of course, before providing the predictors $\mathbf{L}_t(\text{CWV}_t)$ in input to CWV-Net, they are binned by a factor of two and the channels corresponding to wavelengths higher than 925 nm are discarded according to the input format defined in Section III.

B. Loss Function for CWV-Net Training

The power of DNN depends on the connections (weights) of the entire network. These weights are "learned" during the training of the network by minimizing a chosen loss function over the training set pairs. In formulas

$$\boldsymbol{\Omega} = \arg \min_{\tilde{\Omega}} \ \underset{t=1,\ldots,N}{\text{loss}}\left[g_{\text{net}}\left(\mathbf{L}_t, \tilde{\boldsymbol{\Omega}}\right), \text{CWV}_t\right] \tag{4}$$

where, $\boldsymbol{\Omega}$ denotes the vector of the network weights, $g_{\text{net}}(\cdot)$ is the transformation operated by the network on the predictors (inputs), N is the total number of training pairs, and loss[·] is the loss function computed over the entire training set. The loss function adopted to train CWV-Net is based on the mean absolute percentage error (MAPE) defined as

$$\text{MAPE}\left(\tilde{\Omega}\right) = \frac{1}{N}\sum_{t=1}^{N} \frac{\left|g_{\text{net}}\left(\mathbf{L}_t, \tilde{\Omega}\right) - \text{CWV}_t\right|}{\text{CWV}_t} \cdot 100. \tag{5}$$

Typically, CWV has values ranging from 0.5 $\text{g} \cdot \text{cm}^{-2}$ (very dry atmosphere) to 5 $\text{g} \cdot \text{cm}^{-2}$ (very wet atmosphere). MAPE is chosen not to penalize small values of CWV.

It is noted that CWV has positive values (>0.5) and this ensures the absence of division by zero and the nonnegativity of MAPE.

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To limit the risk of overfitting, it is good practice to add a L2 regularization to the loss function (see [40]). Thus

$$\text{loss}(\tilde{\boldsymbol{\Omega}}) = \text{MAPE}(\tilde{\boldsymbol{\Omega}}) + \frac{\alpha}{2}\|\tilde{\boldsymbol{\Omega}}\|^2 \tag{6}$$

where $\|\cdot\|^2$ denotes the L2-norm (Euclidean norm) and α is a free parameter that is tuned empirically (in our case we set $\alpha = 10^{-4}$).

The loss function in (6) is minimized starting from randomly values for the network weights and using the Adam stochastic gradient-based optimizer (see [41]).

V. Results on Simulated Data

In this section, we analyze the performance of the CWV-Net on simulated data. The analysis aims to evaluate the ability of the network of estimating the values of CWV without any knowledge about the atmospheric visibility, the solar zenith angle, and the adjacency effects. The analysis also aims to compare CWV-Net with the LRSWE algorithm and the APDA algorithm, and to show the better flexibility that the DNN approach offers in typical operating conditions.

A. Experiments Description and Algorithms Settings

The analysis is carried out with reference to two scenarios. The first scenario concerns the aerial application. Specifically, we consider the case in which the airborne platform flies at 1 Km. The second scenario is related to a typical satellite application, where we have assumed a flight height of 600 Km.[1] In both the scenarios, we have assumed that the monitored scene is in the Mid-Latitude region and that data are collected during the Summer.

To perform the analysis, two CWV-Net are trained by properly setting the fixed parameters M and h. Specifically, M is set to 2 that corresponds to the MODTRAN mid latitude Summer model, while h is set to 1 and 600 km, respectively in accordance with the two case studies. For the two networks, training is performed over more than 5000 epochs with a decreasing learning rate (it starts from 0.01 and is reduced by 2/3 every 500 epochs).

Two different test sets (one for each case study) are generated using the simulation strategy in Section IV. Test sets are simulated by setting $M = 2$ and h in accordance with the corresponding case study. For each test set, we generated 10^3 surface spectral reflectance vectors $\boldsymbol{\rho}$ and the same number of adjacent spectral reflectance vectors $\boldsymbol{\rho}_a$. They are combined with the radiometric quantities generated by choosing at random the values of the aerosol parameter IHAZE from the values reported in Table II. For the parameters V, ϑ, and CWV all the values in Table III were considered. It is noted that test data differ from the training set for the surface reflectance vectors, the adjacent reflectance vectors, and, also, for the values of the atmospheric parameters V, ϑ, and CWV.

To test the estimation performance with respect to the noise affecting hyperspectral data, noisy at-sensor radiances are simulated. Noise realizations are generated according to the well-established signal dependent noise model proposed in [42] and [43]. They are drawn from a Gaussian distribution with zero mean and diagonal covariance matrix whose kth diagonal element σ_k^2 linearly depends on the at-sensor radiance in the kth sensor channel

$$\sigma_k^2 = \sigma_{\text{ph},k} L(\lambda_k) + \sigma_{\text{th},k}^2 \tag{7}$$

where λ_k is the central wavelength of the kth sensor channel, and $\sigma_{\text{ph},k}$ and $\sigma_{\text{th},k}^2$ account for the photon noise and the thermal noise, respectively. Specifically, we assume that the photon noise and the thermal noise terms have the same statistical power per band, that is $\sigma_{\text{ph},k} L(\lambda_k) = \sigma_{\text{th},k}^2$. $\sigma_{ph,k}$ is modeled according to [42] as

$$\sigma_{\text{ph},k} = \frac{\beta}{\lambda_k \cdot m(\lambda_k)} \tag{8}$$

where $m(\lambda_k)$ is the sensor quantum efficiency and β is a multiplicative factor that allows the desired signal-to-noise ratio (SNR) to be set. SNR is defined as the ratio between the total at sensor radiance power $\sum_{k=1}^{N_b} L^2(\lambda_k)$ and the total noise power $\sum_{k=1}^{N_b} \sigma_k^2$, and can be written as

$$\text{SNR} = \frac{\sum_{k=1}^{N_b} L^2(\lambda_k)}{2\beta \sum_{k=1}^{N_b} \frac{L(\lambda_k)}{\lambda_k \cdot m(\lambda_k)}}. \tag{9}$$

In (9) N_b is the number of sensor channels. Equation (9) makes explicit the dependence of the SNR on the multiplicative factor β. For $m(\lambda_k)$, we have not a specific model, it is approximated as a Gaussian function on the basis of the behavior of $\sigma_{ph,k}$ experienced in real data (see [42]).

As stated in the first part of this section, CWV-Net is compared with the LRSWE algorithm and the APDA algorithm. Both the algorithms require LUTs derived for specific values of the atmospheric visibility and the solar zenith angle. In the following comparative analysis, the LUTs adopted for the two algorithms are derived by setting $V = 35$km and $\vartheta = 0^\circ$.

LRSWE also needs the knowledge of the at-sensor radiance of the adjacent pixels $\boldsymbol{L}_a$. As to $\boldsymbol{L}_\text{a}$, in the derivation of the algorithm, it was assumed that the kth component of the vector is obtained by the RTM in (1) and (2) with $\rho(\lambda_k) = \rho_a(\lambda_k)$

$$\begin{aligned} \boldsymbol{L}_a(\lambda_k) &= L_{\text{GND}}(\lambda_k) + L_p(\lambda_k) \\ L_{\text{GND}}(\lambda_k) &= \frac{E_g(\lambda_k)}{\pi} \frac{\tau_1(\lambda_k) + \tau_2(\lambda_k)}{1 - S(\lambda_k)\rho_a(\lambda_k)} \rho_a(\lambda_k) \end{aligned} \tag{10}$$

where, for simplicity, the dependence of the radiometric quantities on the atmospheric parameters has been omitted.

TABLE III
Variable Atmospheric Parameters Adopted in Simulating the Test Sets

Atmospheric Parameters in the test sets	
$\vartheta(deg)$	0, 5:10:45
$V(Km)$	15:10:65, 100
$CWV\ (^{g}/_{cm^2})$	0.5:0.25:5

[1]It is worth noting that, the effect of the atmosphere on the radiative transfer in VNIR spectral range are actually negligible for altitude greater than 100 Km.

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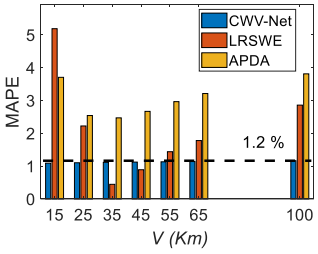


Fig. 3. $h = 1$ Km. Estimation performance in terms of MAPE of CWV-Net, LRSWE, and APDA for different values of V and obtained in the absence of noise and with $\vartheta = 0°$.

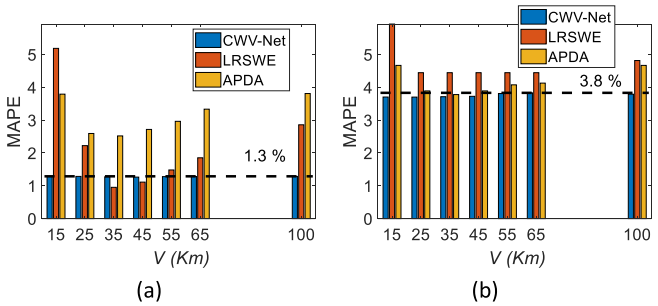


Fig. 4. $h = 1$ Km. Estimation performance in terms of MAPE of CWV-Net, LRSWE, and APDA for different values of V, with $\vartheta = 0°$ and with (a) SNR = 50 dB and (b) SNR = 35 dB.

In general, $\boldsymbol{L}_a$ is obtained as a linear combination of the radiances of a neighborhood of the pixel under consideration. In this analysis, using the simulation strategy in Section IV (see Fig. 2), we generate also the vector $\boldsymbol{L}_a$ that is used as an input together with $\boldsymbol{L}$ for the LRSWE algorithm. Therefore, the algorithm is tested in the ideal condition in which there is no uncertainty on the value of $\boldsymbol{L}_a$.

It is noted that both LRSWE and APDA work on a specific water absorption band. In the following analysis, we consider the water absorption band around 820 nm. Such choice is made to avoid the bias due to leaf water content generally experienced over vegetated pixels.

As to LRSWE the low-rank subspace reflectance matrix in the selected absorption band is derived from the library $\boldsymbol{D}$, by the singular value decomposition (SVD)-based procedure described in [16].

When applying both the algorithms, the data are binned with a factor of two and in the case of CWV-Net the bands with central wavelength greater than 925 nm are discarded.

Performance are evaluated in terms of MAPE between the estimation provided by each algorithm and the value of CWV with which the data are simulated.

B. Aerial Scenario: Results and Discussion

Fig. 3, which refers to a flight height of 1 km, shows the values of MAPE obtained for all the CWV values in Table III by varying V (according to Table III). The figure was obtained in the absence of noise and with the same a solar zenith angle of 0° adopted to derive the LUTs for both LRSWE and APDA. The total number of test data generated in this case study is 6.65×10^5.

Fig. 3 shows that CWV-Net has good estimation performance with values of MAPE in the order of 1.2% and that CWV-Net has approximately the same estimation performance regardless of the value of V. In particular, results in Fig. 3 show that CWV-Net provides good estimation performance without any knowledge of the adjacent at sensor radiance and of the atmospheric visibility value.

As expected, this conclusion cannot be extended to the LRSWE algorithm. It outperforms CWV-Net providing the lowest value of MAPE (0.44%) for $V = 35$ km, which is the visibility value adopted in the derivation of the LUT. However, the performance degrades when the atmospheric visibility deviates from the hypothesized value. LRSWE is outperformed by CWV-Net for all the considered V values with the exception of $V = 35$ km and $V = 45$ km.

The APDA algorithm provides performance worse than that attained by CWV-Net. Fig. 3 also shows that APDA has the best performance (MAPE $= 2.4\%$) for the visibility value adopted to derive the LUT ($V = 35$ km). The performance worsens when the atmospheric visibility of the test data deviates from that value.

Furthermore, Fig. 3 confirms the conclusion drawn in [16] i.e., LRWSE generally outperforms APDA. Fig. 4(a) and (b) show the results obtained in presence of noise with SNR equal to 50 and 35 dB, respectively.

It is noted that, even in the presence of noise, CWV-Net provides the same estimation performance for all the values of the parameter V. They also show that CWV-Net performance degrades with decreasing SNR. Specifically, for CWV-Net, values of MAPE are below by 1.3% for SNR = 50 dB and 3.8% for SNR = 35 dB. Comparing the results provided by CWV-Net with those of LRSWE, we can see that the DNN-based algorithm is more robust to noise. In fact, the increase in MAPE values with decreasing SNR is more significant for LRSWE than for CWV-Net. In the ideal case of $V = 35$ km and $\vartheta = 0^{\circ}$, LRSWE yields MAPE values equal to 0.44% in the absence of noise, 0.9% with SNR = 50 dB, and 4.4% with SNR = 35 dB. Similar conclusions can be drawn by comparing CWV-Net with APDA. It is worth noting that, also in the presence of noise, APDA performance degrades when test data visibility deviates from the value adopted to derive the LUT.

In Fig. 5(a)–(c), we test the performance of the three algorithms for the values of the solar zenith angle in Table III, and assuming $V = 35$ km. The total number of test data generated is 6.65×10^5. Fig. 5(a) refers to the case of noise free data, while Fig. 5(b) and (c) show the results obtained with SNR equal to 50 and 35 dB, respectively.

Here again, in the absence of noise, CWV-Net provides very low values of MAPE, in the order of 1.1%. Furthermore, Fig. 5(a) shows that the performance of CWV-Net is approximately the same for all the values of ϑ. This is confirmed also in the noisy cases where, of course, the performance degrades with decreasing SNR. With high SNR (50 dB), CWV-Net yields very low values of MAPE (1.3%). The error index grows up to 3.7% when SNR decreases to 35 dB.

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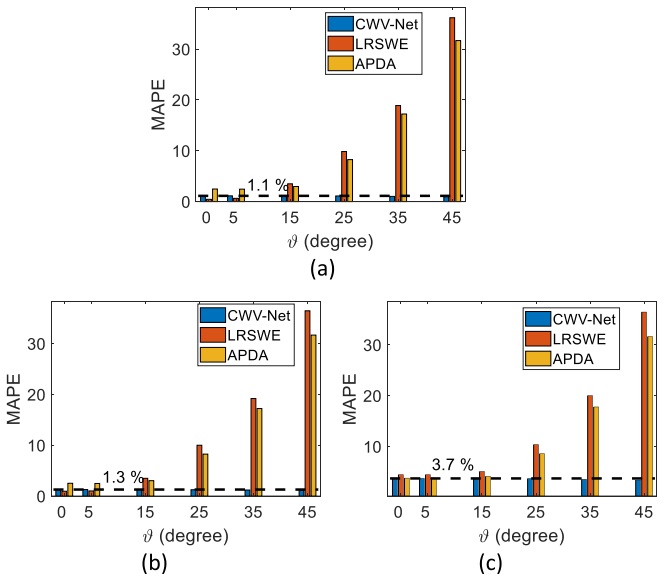


Fig. 5. $h = 1$ Km. Estimation performance in terms of MAPE of CWV-Net, LRSWE, and APDA for different values of ϑ, obtained with $V = 35$ Km. (a) Absence of noise. (b) SNR = 50 dB. (c) SNR = 35 dB.

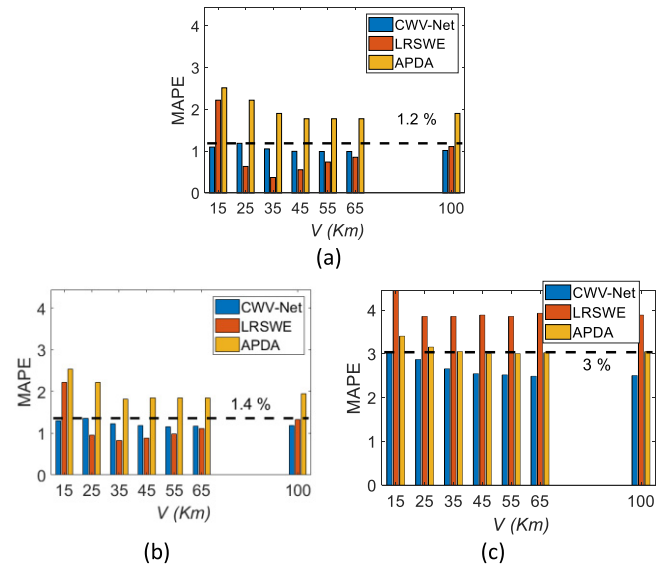


Fig. 6. Satellite scenario. CWV-Net, LRSWE, and APDA estimation performance in terms of MAPE for different values of V, obtained with $\vartheta = 0°$. (a) Absence of noise. (b) SNR = 50 dB. (c) SNR = 35 dB.

As opposite to CWV-Net, the performance of both LRSWE and APDA strongly depends on the values of ϑ. Particularly, the estimation error increases as ϑ deviates from the value ($\vartheta = 0°$) used to compute the LUTs adopted by the two algorithms. To give some numerical examples, in the case of LRWSE and in the absence of noise, MAPE goes from 0.44% in the ideal case ($V = 35$ km and $\vartheta = 0°$) to 36% when $\vartheta = 45°$. In the case of APDA (in the absence of noise), MAPE has its minimum value (2.4%) in the ideal case and grows up to 31% when $\vartheta = 45°$.

It is worth noting that we obtained similar results by considering flight heights of 2 and 3 km.

C. Satellite Scenario: Results and Discussion

Experiments similar to those presented in Section V-B are carried out in the case of a satellite scenario, where the contribution of CWV on the at-sensor radiance is more significant than in the aerial scenario as the electromagnetic radiation follows a longer path upward.

Fig. 6(a)–(c) show MAPE values for the three algorithms obtained by letting ϑ to 0 and by varying V and the contribution of noise. In the absence of noise [see Fig. 6(a)], CWV-Net has approximately the same performance for all values of V.

Specifically, it provides values of MAPE lower than 1.2%. The performance of LRSWE varies with V. The best result (MAPE = 0.37%) is obtained in the ideal case of $V = 35$Km. However, in this case, LRSWE outperforms CWV-Net for all values of V except for $V = 15$ km and $V = 100$ km. APDA attains the best performance (MAPE = 1.7%) for the value of V equal to that adopted to derive the LUT ($V = 35$ Km). Its performance degrades slightly for $V > 35$ Km. The worst performance is obtained for $V = 15$ Km. Results confirm that the LRSWE performs better than APDA (see [16]).

In presence of moderate noise contribution [SNR = 50 dB, see Fig. 6(b)], the performance of CWV-Net remains constant

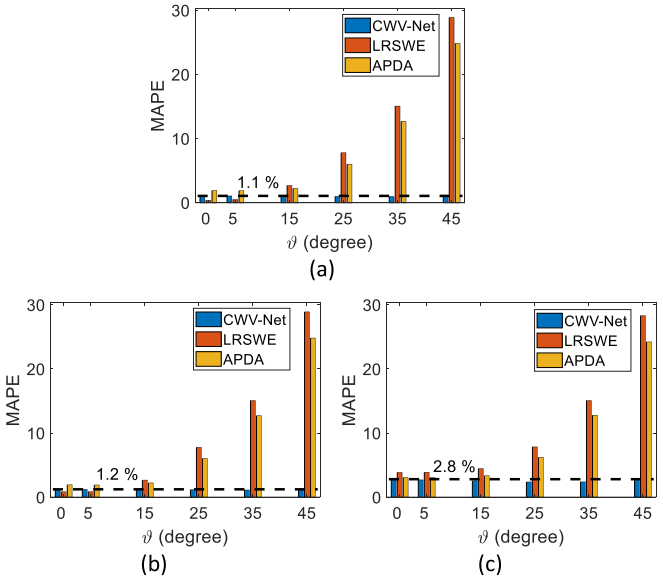


Fig. 7. Satellite scenario. CWV-Net, LRSWE, and APDA estimation performance in terms of MAPE for different values of ϑ, obtained with $V = 35$ Km. (a) Absence of noise. (b) SNR = 50 dB. (c)SNR = 35 dB.

(MAPE in the order of 1.4%) with respect to V and are similar to that provided by LRSWE. When SNR drops to 35dB [see Fig. 6(c)], CWV-Net outperforms LRSWE by providing values of MAPE lower than 3%. APDA seems to be more robust to noise with respect to LRSWE. However, APDA provides lower accuracy than CWV-Net in all the considered cases.

Setting $V = 35$ km and varying ϑ, the three algorithms yield the performance shown in Fig. 7(a)–(c). Results show the robustness of CWV-Net with respect to ϑ in all the cases of noise absent, SNR = 50 and SNR = 35 dB. Specifically, CWV-Net has very low values of MAPE ($\leq$ 1.2%) with moderate [see Fig. 7(b)] or absent noise [see Fig. 7(a)]. MAPE increases (while remaining below 3%) when SNR drops to

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35 dB [see Fig. 7(c)]. On the contrary, the performance of LRSWE degrades when the difference between the values of ϑ adopted in the LUT and that of the observed data increases. This behavior is confirmed for all the cases considered in our experiment and is observed also for the APDA algorithm.

We would like to remark again that in all the experiments discussed in this section, LRSWE has been tested with no uncertainty on the value of $\boldsymbol{L}_a$. This is an ideal condition that generally does not occur in real applications where $\boldsymbol{L}_a$ must be estimated from the analyzed image by assuming an atmospheric PSF model.

VI. Example Over Real Data

In this section, we present the results obtained by applying CWV-Net to a real hyperspectral image. It important to noted that evaluating the performance of any CWV retrieval algorithm on real data is impractical. Indeed, performance analysis would require a great amount of hyperspectral data including precise measurements of the parameter CWV that varies spatially over the scene. Unfortunately, there are no data set available that meet these requirements. Therefore, simulation remains the only way to perform accurate performance analysis and comparison, and to draw reliable conclusions.

The lack of information about the CWV values at the time of the acquisition and the uncertainty on the radiometric calibration of the data prevents the assessment and the comparison of the performance provided by CWV-Net, the LRSWE algorithm, and the APDA algorithm over the selected data set. The example reported in the following provides a proof of concept aimed at showing that CWV-Net produces plausible results over real data.

The hyperspectral image adopted in our example is part of the data set acquired during the "Viareggio 2013 trial" that was held in Viareggio (Italy), on May 8 and 9, 2013 (see [44]). All the data were collected by the SIM.GA sensor introduced in Section III. For this experiment, we focus on the image named D2F22H2 that was acquired on May 9, 2013 under clear sky conditions over a scene including a parking lot (where a number of man-made objects were deployed), several sport facilities buildings, and a football field. The flight height was of about 1.2 km that corresponds to a nominal ground spatial resolution of about 0.6 m.

The image is equipped with ground truth data including the map of the deployed man-made objects (targets) and their spectral reflectances measured by an ASD FieldSpec hand-held spectroradiometer. In Fig. 8, we show the red green blue (RGB) representation of the image and, superimposed, the locations of the man-made objects. Each object is identified by a label.

For this example, CWV-Net was trained according to the procedure described in Section IV. Specifically, the training set was generated by setting $h = 1.2$ km and the MODTRAN parameter M to 2 corresponding to mid-latitude Summer. This choice is motivated by the geographical coordinates of the scene and by the fact that acquisition was made in the late Spring season.

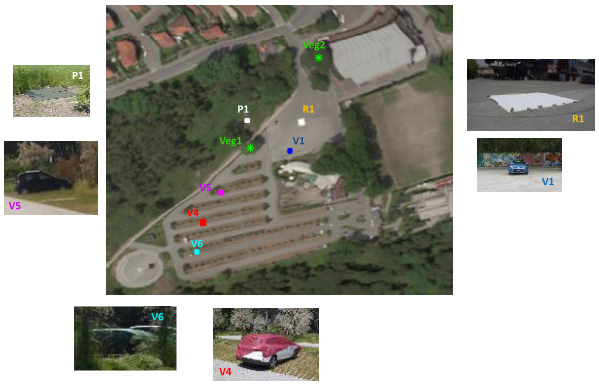


Fig. 8. RGB representations of the hyperspectral image adopted in the example.

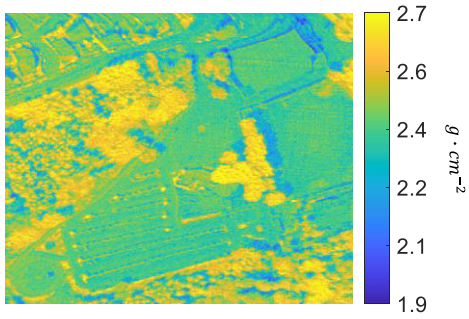


Fig. 9. CWV estimates obtained by applying CWV-Net to each pixel of the image.

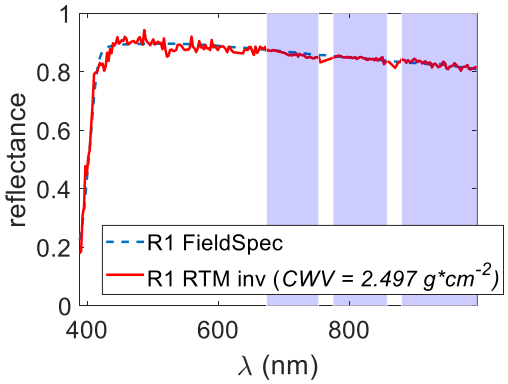


Fig. 10. Target R1. Comparison between the reflectance spectrum measured by the FieldSpec instrument and that obtained from the image by inverting the RTM and using the CWV estimate provided by CWV-Net.

The CWV values expressed in $\mathrm{g} \cdot \mathrm{cm}^{-2}$ provided by the network for each pixel of the image are reported in the map of Fig. 9.

Comparing the CWV map in Fig. 9 with the RGB image in Fig. 8, we can note that the lowest values of CWV are obtained in the shadowed areas. The proposed algorithm does not provide reliable results in those areas because the adopted RTM does not account for the shadow effects. In the remaining pixels of the image, CWV-Net provides values in the range $[2.3, 2.7]$ $\mathrm{g} \cdot \mathrm{cm}^{-2}$. Specifically, the mean value of the estimated CWV is 2.46 $\mathrm{g} \cdot \mathrm{cm}^{-2}$ and its standard deviation is 9×10^{-2} $\mathrm{g} \cdot \mathrm{cm}^{-2}$. The obtained values are compliant with those that characterize, at least on a large scale, the region of the world that includes the monitored site.

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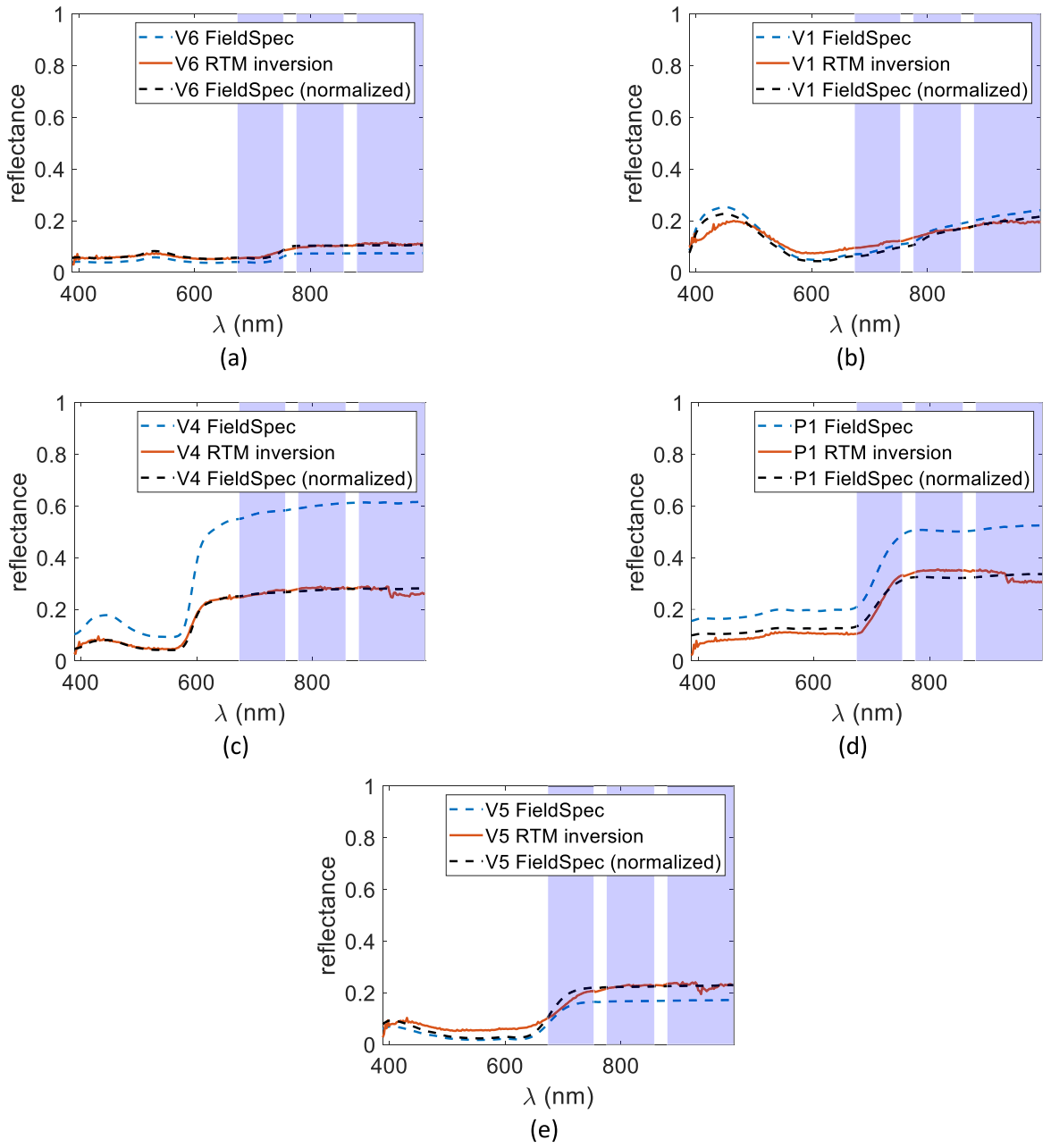


Fig. 13. Comparison between the reflectance spectrum measured by the FieldSpec instrument, its scaled version and that obtained from the image by inverting the RTM for the ground truth targets. (a) *V6*. (b) *V1*. (c) *V4*. (d) *P1*. (e) *V5*.

As already stated, in the absence of CWV real value measurements, it is not possible to carry out a quantitative assessment of the algorithm's performance. However, one can obtain a sort of indirect validation by analyzing the reflectance spectrum obtained inverting the RTM for some of the image pixels and making use of the values of CWV estimated by CWV-Net.

To invert the RTM (AC), we need the values of the other relevant MODTRAN parameters and specifically: ϑ, V, and the aerosol model defined by the parameter IHAZE. The value of ϑ is obtained considering the geographical coordinates of the monitored scene, and the acquisition date and time. While the values of IHAZE and V are estimated by using the Boundary Shaded Pixels based algorithm proposed in [9]. This latter provides $V = 33$ km and IHAZE $= 4$ corresponding to the "Maritime" aerosol type.

AC is applied to several pixels of the image. In Fig. 10, we show the results (red plot) obtained for the target labeled as R1 (see Fig. 8). It is a white calibration tarp that was adopted for its isotropic reflective behavior that makes it a good approximation of a Lambertian surface. Fig. 10 also shows the spectral reflectance of R1 measured by the FieldSpec instrument and included in the ground truth metadata. The transparent blue patches highlight the water absorption bands. The high similarity between the atmospheric compensated spectrum and the measured one in those bands,

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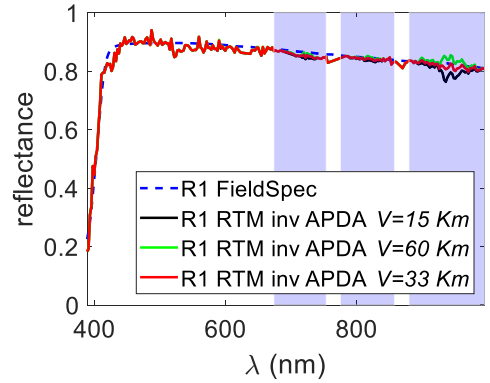


Fig. 11. Target R1. Comparison between the reflectance spectrum measured by the FieldSpec instrument and that obtained from the image by inverting the RTM and using the CWV estimates provided by APDA in the 820-nm absorption band. CWV estimates for RTM inversion are obtained using the LUT derived with the correct visibility value (red line) and the LUTs derived with incorrect visibility values (green and black lines).

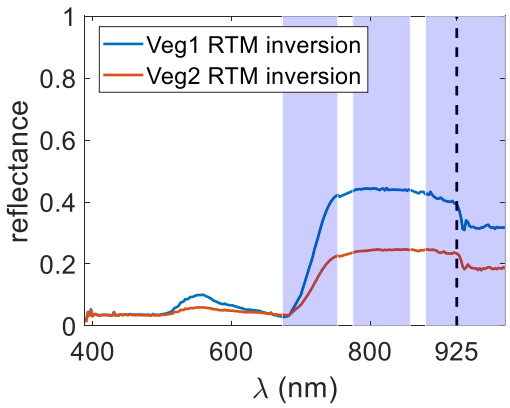


Fig. 12. Reflectance spectra of the pixels labeled as Veg1 and Veg2 obtained by inverting the RTM.

shows that CWV value provided by CWV-Net allows the water absorption to be correctly compensated. It is noted that for completeness, in Fig. 10, we report the value of CWV (2.497 g $\cdot$ cm^{-2}) provided by CWV-Net in the pixel occupied by R1.

To give an example, we applied also the APDA algorithm to the real image presented in this section by considering the water absorption band around 820 nm. In Fig. 11, we plot the results of the AC on the target R1 obtained using the CWV estimates provided by the APDA algorithm in three cases. In the first case (red line), APDA is applied by using the LUT derived with the correct visibility value ($V = 33$ Km). In this case, we obtain a result very similar to that provided by CWV-Net that does not use any *a priori* knowledge of either the solar zenith angle or the atmospheric visibility. Specifically, the estimated value of CWV (2.5 g $\cdot$ cm^{-2}) is very close to that provided by CWV-Net (2.497 g $\cdot$ cm^{-2}).

In the other two cases, we used the LUT derived with incorrect values of V. Specifically, the green plot is obtained for $V = 60$ km and the black line is obtained for $V = 15$ Km. The CWV estimates provided by the APDA algorithm in the two cases are 2.6 and 2.4 g $\cdot$ cm^{-2}, respectively. It is noted that in these two cases the retrieved reflectance spectra nearby 950 nm, differ from the spectral reflectance measured by the FieldSpec. This can be considered as an indirect proof of the errors induced on the APDA estimates by the incorrect visibility value adopted in deriving the LUT.

As another example of the results provided by the proposed CWV-Net, in Fig. 12, we show the reflectance spectra obtained by inverting the RTM for two pixels of vegetation, those labeled as Veg1 and Veg2 in Fig. 8. For those pixels, the FieldSpec measurements are not available. However, it is interesting to note how the water absorption features (blue patches) are correctly compensated. It is worth noting that both the spectra have the classical behavior of the canopies where the absorption feature in the spectral region above 925 nm is due to the leaf water content.

For completeness, in Fig. 13(a)–(e), we compare the reflectance spectra of the other targets included in the ground truth with those obtained from the corresponding image pixels by inverting the RTM and using the CWV estimates provided by the proposed algorithm. It is noted that neither the adopted RTM nor the ASD FieldSpec account for the bidirectional reflectance distribution function (BRDF, see [45]), whose effect is generally roughly approximated as a scaling factor. For this reason, in each figure, we also plot the version of the measured FieldSpec spectrum scaled so as to have the same energy of the image-retrieved spectrum. In all the figures, we can see that the water absorption is well compensated.

VII. Conclusion

In this article, CWV-Net, a 1D-CNN-based DNN, is proposed for the retrieval of the total CWV content of the atmosphere from VNIR hyperspectral images. CWV-Net works on each image pixel to account for the spatial variability of the parameter CWV within the observed scene. The network is specifically designed and trained to works without the knowledge of other significant atmospheric model parameters, such as V, ϑ, and the atmospheric PSF.

The network training involves synthetic data that are generated according to a well-established RTM, using the MODTRAN radiative-transfer code and available spectral reflectance libraries.

Results on simulated data have shown the potential and flexibility of the DNN-based approach in estimating CWV independently of the values of V and, ϑ and regardless of the adjacency effects. Specifically, results obtained in both aerial and satellite scenarios showed that in the presence of moderate noise (SNR $=$ 50 dB), CWV-Net has estimation error in the order of 1%. Such an error increases when SNR decreases. However, for SNR $=$ 35 dB the error remains less than 4%. The application of CWV-Net to a real hyperspectral image has shown that the DNN-based approach provides reasonable results even outside the controlled environment of the simulated data.

It is worth noting that the DNN-based approach offers an additional advantage concerning the computational load.

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Although training is generally computationally expensive, applying CWV-Net to a given image requires a low-computational load that makes the proposed approach suitable for real-time applications. Just to give an example, using a MATLAB implementation on a personal computer equipped with Intel Core i7-4820K CPU at 3.7 GHz and with a 16 GB of RAM, we experienced a computing time of about 7 μs per pixel (228 elements vector). To process the real image of Section VI having 450 $\times$ 375 pixels, our MATLAB implementation of CWV-Net took a total time of 1.2 s. Processing times can be significantly reduced using dedicated hardware or modern graphical processing units.

We would like to conclude by noting that the CNN architectures generally provide a certain degree of invariance with respect to input nodes' shift, which could be useful for obtaining the robustness of the algorithm with respect to spectral miscalibration. This issue requires further investigation and is one of the objectives of our ongoing activities. Future activities will also concern the analysis of the impact of the radiometric miscalibration on the performance of CWV-Net and the study of possible methods to reduce this impact.

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